UGANDA CHRISTIAN UNIVERSITY

OBJECT-ORIENTED PROGRAMMING USING PYTHON

GROUP ASSIGNMENT: INTRODUCTION TO DJANGO WEB DEVELOPMENT

Assignment type	Group research, presentation and practical demonstration
Presentation date	Thursday, 5 November 2026

1. Assignment Brief

In your existing project groups, independently study Django, a Python web development framework, and prepare a classroom presentation supported by a working practical demonstration. The aim is to connect the Python and object-oriented programming concepts studied so far to the development of web applications.

2. Areas to Cover

1. Introduction to Django: its purpose, key features, and examples of applications built with it.

2. Django architecture: explain the Model–View–Template (MVT) pattern and how requests are processed.

3. Development environment: Python virtual environment, Django installation, and creation of a project and app.

4. Project structure: explain the roles of manage.py, settings.py, urls.py, views.py, models.py and templates.

5. URLs, views and templates: demonstrate how a browser request produces a rendered web page.

6. Models and databases: explain how Django models represent data and introduce migrations and the default SQLite database.

7. Object-oriented concepts in Django: identify where classes, objects, inheritance and encapsulation appear in the framework.

3. Required Practical Demonstration

Each group must build and demonstrate a small Django web application, preferably a Book Catalogue or another simple system of its choice. The demonstration must show:

• Creating and running a Django project and application.

• Configuring a URL and a view that displays a page through a template.

• Defining at least one Django model (e.g., Book with title, author and availability).

• Creating and applying migrations, then adding and displaying sample records from SQLite.

• Explaining the connection between the Django model and the Python OOP concepts already covered.

4. Expected Deliverables

• A group presentation (approximately 10–12 slides) explaining the assigned concepts.

• A working Django project demonstrated live in class, with at least three sample records.

• The project source code and a brief README containing installation and execution steps.

• Participation by all group members in the presentation or demonstration.

5. Presentation and Assessment

Each group will have approximately 10 minutes for the presentation and practical demonstration, followed by 5 minutes of questions. Be prepared to explain your code rather than only display screenshots.

Assessment criterion	Marks
Accuracy and clarity of Django concepts	25
Working practical demonstration	40
Application of Python/OOP concepts	15
Presentation organization and teamwork	10
Responses to questions	10
Total	100